What is Science?

Comparative Politics

POLC-2300-01

August 26, 2021

Last time

Comparative politics is a subfield of **Political Science**

Some philosophers may think this is an oxymoron:

1. **Politics:** Normative opinions
2. **Science:** True, unquestionable knowledge

Can there be a **SCIENCE** of **POLITICS**?

Today: Focus on the **Science** part

Tuesday: Focus on **Politics**

Jamboard

The goal of science

- **Inference:** Using `what we know` to learn about `what we do not know`
- CGG: *"Science is the quest for knowledge that relies on criticism"*
- Science embraces the idea that we can be wrong, even if we seem to be right now
- **Scientific statements** are **falsifiable**

Definition

A *scientific statement* is:

A `meaningful declarative` sentence that is `unambiguously true or false`

- **Meaningful:** We agree on its interpretation
- **Declarative:** Claims or asserts something
- **Unambiguously true or false:** One or the other, but not both, not neither

Unscientific statements

Political Science is the scientific study of politics

Strong states overcome special interests to implement policies that are best for the nation

I feel good today

God created the world `vs.` *Edison created the light bulb*

Aside

Scientific statements are true or false

Scientific knowledge/evidence is neither

Why?

How do you **evaluate** a scientific statement?

The Scientific Method

- **Step 1:** Question
- **Step 2:** Theory/Model
- **Step 3:** Implications/Hypotheses
- **Step 4:** Observation/Testing
- **Step 5:** Evaluation

An exercise

Study by Tulane researcher suggests marijuana can cause infertility in men

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Cannabis has compounds that can affect the male reproductive system, yet prior studies haven't agreed on how that impacts fertility. Photo by Shutterstock.

An exercise

1. **Question:** Does marijuana cause infertility?
2. **Theory:** THC kills cells
3. **Implication:** Male smokers have lower semen volume than non-smokers
4. **Observation:** Bring men to lab, measure sperm, findings go along with expectations
5. **Evaluation:** "Study by Tulane researcher `suggests` marijuana `can` cause infertility in men"

Testing theories

- Can't test a theory because it's a **statement**
- Need to **believe** (or **pretend**) a theory is true to engage with it scientifically
- Instead, we test hypotheses, implications, arguments that **follow from the theory**
- **Argument:** Logically connected statement
- **Valid argument:** Accepting premises compels us to accept conclusion
- **Invalid argument:** Accepting premises frees us from accepting/rejecting conclusion

Example 1

Premise

Marijuana causes infertility

Observation

Person consumes marijuana

Conclusion

Person is infertile

VALID

Example 2

Premise

Marijuana causes infertility

Observation

Person does not consume marijuana

Conclusion

Person is not infertile

INVALID

Example 3

Premise	Marijuana causes infertility among men
Obs. 1	Person consumes marijuana
Obs. 2	Person is a woman
Conclusion	Person is fertile

INVALID

Scientific challenges in CP

We want to test:

- `Marijuana causes infertility`

But we can only test:

- `Marijuana causes infertility among men`
- `Marijuana causes infertility among women`
- `Smoking marijuana causes infertility among men who go to a clinic to consult about potential fertility problems` (very specific!)

Scientific critiques

Valid critiques to **valid arguments** fall in **three categories:**

1. Omitted variable bias:
   - `Stress causes men both to smoke marijuana and to become infertile`
2. Reverse causation:
   - `Infertility causes men to smoke marijuana`
3. Selection bias:
   - `Men who smoke marijuana wait longer before going to the doctor`

Valid critiques

In any case, a valid critique implies:

1. Observed implications also follow from **alternative theories**
2. **Cannot falsify** the theory
3. Need to identify **better implications**

So...

- When things go well, your theory is ~~**accepted**~~ **not rejected**
- When things go wrong, your theory is ~~**rejected**~~ **not accepted**
- But future research can always flip things around!

Takeaways

1. Science differs from other forms of knowledge generation in that it embraces uncertainty
2. Scientific statements/arguments are falsifiable
3. We can only test a few implications of an theory at a time
4. Findings can align with many alternative theories
5. Good science tries really hard to prove itself wrong

Next week

Tuesday: What is politics?

Thursday: Origins of the state